

Review

Definition 1 Let $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a function and $p \in \mathbb{R}^m$. We say that f is differentiable at p if for any there exists a best linear approximation $(Df)_p$ of f at p meaning:

$$f(p+v) = f(p) + (Df)_p \circ v + R_p(v)$$

Where $(Df)_p : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a linear function and R_p is of “higher order”, meaning that:

$$\lim_{||v|| \rightarrow 0} \frac{||R_p(v)||}{||v||} = 0$$

We call $(Df)_p$ the derivative at p (or the Jacobian).

Theorem 1 Let $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be differentiable at p . Let $v \in \mathbb{R}^m$. Introduce the following notation:

$$(Df)_p v = \frac{\partial f}{\partial \vec{v}}$$

Then the following is true:

$$\frac{\partial \vec{f}}{\partial \vec{v}} = \lim_{t \rightarrow 0} \left(\frac{f_1(p+tv) - f_1(p)}{t}, \dots, \frac{f_i(p+tv) - f_i(p)}{t}, \dots, \frac{f_n(p+tv) - f_n(p)}{t} \right)$$

Notation 1 Let $e_1, \dots, e_m$ be the standard basis. Let $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be some differentiable function. We then denote:

$$\frac{\partial f}{\partial x_j} = \frac{\partial f}{\partial e_j}$$

Theorem 2 Let $g : \mathbb{R}^m \rightarrow \mathbb{R}^k$ and $f : \mathbb{R}^k \rightarrow \mathbb{R}^n$. Suppose g is differentiable at $p \in \mathbb{R}^m$ and f is differentiable at $g(p)$. Then $f \circ g$ is differentiable at p with derivative:

$$(D(f \circ g))_p = (Df)_{g(p)} \circ (Dg)_p$$

This is called “chain rule”.

Theorem 3 Let V be some open set of $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function. Let U be some other open set of $\mathbb{R}^n$ and $\varphi : U \rightarrow V$ be some diffeomorphism (i.e. smooth invertible map with smooth inverse). Then the following holds:

$$\int_V f = \int_U (f \circ \varphi) \cdot |\det D\varphi|$$

Suggested Exercises

- I) Let $\gamma : \mathbb{R} \rightarrow \mathbb{R}^n$ be some smooth curve. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function. Notice that $f \circ \gamma : \mathbb{R} \rightarrow \mathbb{R}$ is a single variable function. Show that:

$$\frac{\partial f \circ \gamma}{\partial t} = \frac{\partial f}{\partial x_1} \Big|_{\gamma(t)} \cdot \frac{\partial \gamma_1}{\partial t} \Big|_t + \dots + \frac{\partial f}{\partial x_j} \Big|_{\gamma(t)} \cdot \frac{\partial \gamma_j}{\partial t} \Big|_t + \dots + \frac{\partial f}{\partial x_n} \Big|_{\gamma(t)} \cdot \frac{\partial \gamma_n}{\partial t} \Big|_t$$

- II) Compute the Jacobians of the following functions:

$$\begin{aligned} f(x, y) &= x^2 + y^2 & f(x, y) &= (x + y)^2 \\ \gamma(t) &= (\cos(t), \sin(t)) & \gamma(t) &= (t, e^t) \\ g(r, \varphi) &= (r \cos \varphi, r \sin \varphi) \end{aligned}$$

- III) Compute the volume of a sphere of radius r . (*Hint: Do a change of variables to spherical coordinates: $g(r, \theta, \varphi) = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$*)

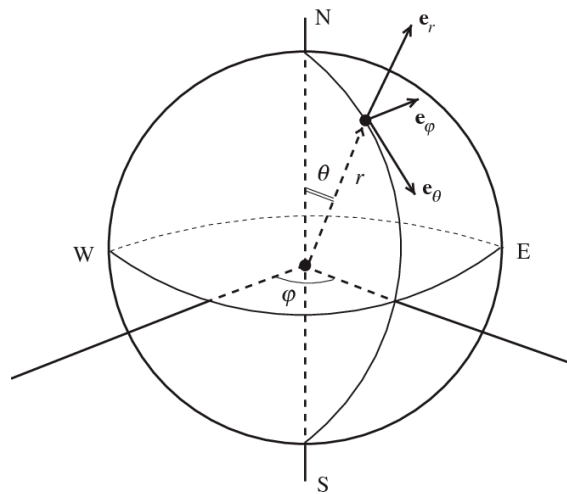


Figure 1: Copied from google images.